BASE CHANGE FOR THE DERIVED PUSHFORWARD OF THE INTERNAL HOM OUT OF A PERFECT OR PSEUDO-COHERENT OBJECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The single-complex base change lemma and its condition-inheritance corollary reduce base change assertions for maps into internal-Hom pushforwards to the verification of the equivalent conditions for a canonical comparison map. invoked at proof lines 6963-6968. Reduces the assertion to the underived pullback comparison map.
- The canonical map from the derived pullback to the underived pullback of the flat complex G satisfies the base change conditions, as shown in the proof of the tensor base change lemma using the flatness of the terms. invoked at proof lines 6971-6972. Concludes the base change isomorphism.

Source reference. Original lemma 0DJ9: sources/stacks-perfect/pinned-original-source.tex, source lines 6749–6749. Cited at proof lines 6963-6964 to reduce the assertion; the condition verification is inside the counted span.

Source reference. Original lemma 0DJA: sources/stacks-perfect/pinned-original-source.tex, source lines 6823–6823. Cited at proof lines 6964-6965 to transfer the criterion to the internal Hom; the reduction is inside the counted span.

Source reference. Original lemma 0DJ8: sources/stacks-perfect/pinned-original-source.tex, source lines 6654–6654. Cited at proof lines 6968-6970 as the condition to be verified for the comparison map; the verification citation is inside the counted span.

Source reference. Original lemma 0A1D: sources/stacks-perfect/pinned-original-source.tex, source lines 6880–6880. Cited at proof lines 6971-6972 for the already-done verification of the conditions; the conclusion of the lemma is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a quasi-compact and quasi-separated morphism of schemes. Let E be an object of $D(\mathcal{O}_X)$. Let $\mathcal{G}^\bullet$ be a complex of quasi-coherent $\mathcal{O}_X$-modules. If*

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(1) *E is perfect, $\mathcal{G}^\bullet$ is a bounded above, and $\mathcal{G}^n$ is flat over S, or*
(2) *E is pseudo-coherent, $\mathcal{G}^\bullet$ is bounded, and $\mathcal{G}^n$ is flat over S,*

then formation of

$$Rf_*R\mathcal{H}om(E, \mathcal{G}^\bullet)$$

commutes with arbitrary base change (see proof for precise statement).

Proof. The statement means the following. Let $g : S' \to S$ be a morphism of schemes and consider the base change diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

in other words $X' = S' \times_S X$. The lemma asserts that

$$Lg^*Rf_*R\mathcal{H}om(E, \mathcal{G}^\bullet) \longrightarrow R(f')_*R\mathcal{H}om(L(g')^*E, (g')^*\mathcal{G}^\bullet)$$

is an isomorphism. Observe that on the right hand side we do **not** use the derived pullback on $\mathcal{G}^\bullet$. To prove this, we apply Lemmas 0DJ9 and 0DJA to see that it suffices to prove the canonical map

$$L(g')^*\mathcal{G}^\bullet \to (g')^*\mathcal{G}^\bullet$$

satisfies the equivalent conditions of Lemma 0DJ8. This was shown in the proof of Lemma 0A1D. □